

Week 10

Optimal capital structure

This week will cover the following concepts:

Impacts of financial leverage on return

Earnings Per Share analysis

Learning Objectives

At the end of this module you will:

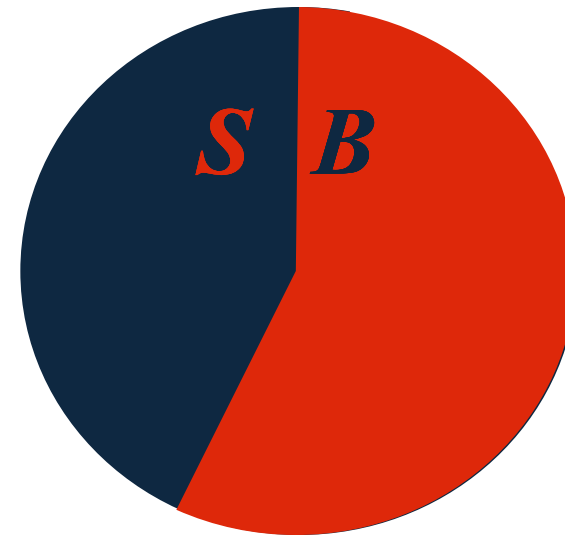
SLO5: assess a firm's financing options and propose a strategy for raising capital

Capital Structure and the Pie

- How should a firm choose its debt-equity ratio? using **pie model**.
- The value of a firm is defined to be the sum of the value of the firm's debt and the firm's equity.

$$V = B + S$$

- If the goal of the firm's management is to make the firm as valuable as possible, then the firm should pick the debt-equity ratio that makes the pie as big as possible.



Value of the Firm

Stockholder Interests

There are two important questions:

1. Why should the stockholders care about maximizing *firm* value?

Perhaps they should be interested in strategies that maximize *shareholder* value.

2. What is the ratio of debt-to-equity that maximizes the shareholder's value?

Maximizing firm value versus stockholder interests

- Suppose the market value of the J. J. Sprint Company is \$1,000. The company currently has no debt, and each of J. J. Sprint's 100 shares of stock sells for \$10. A company such as J. J. Sprint with no debt is called an *unlevered company*.
- Further suppose that J. J. Sprint plans to borrow \$500 and pay the \$500 proceeds to shareholders as an extra cash dividend of \$5 per share. After the issuance of debt, the firm becomes *levered*.
- *The investments of the firm will not change as a result of this transaction.*
- What will the value of the firm be after the proposed restructuring?

Maximizing firm value versus stockholder interests

- The original capital structure and three possibilities under the new capital structure are presented as following:

No debt (original capital structure)		value of debt plus equity after payment of dividend(three possibilities)		
		1	2	3
Debt	\$ 0	\$500	\$500	\$500
Equity	<u>1,000</u>	<u>750</u>	<u>500</u>	<u>250</u>
Firm value	\$1,000	\$1,250	\$1,000	\$750

Maximizing firm value versus stockholder interests

•	Payoff to shareholders		
•	After restructuring		
•	1	2	3
• Capital gains	-\$250	-\$500	-\$750
• dividends	<u>500</u>	<u>500</u>	<u>500</u>
• Net gain or loss	\$250	\$0	- \$250
• To stockholders			

Maximizing firm value versus stockholder interests

- **Conclusions:**

- As it turns out, changes in capital structure benefit the stockholders *if and only if* the value of the firm increases.
- Managers should choose the capital structure that they believe will have the highest firm value, because this capital structure will be most beneficial to the firm's stockholders .

Financial Leverage, EPS, and ROE

Consider an all-equity firm that is considering going into debt. (Maybe some of the original shareholders want to cash out.)

	<u>Current</u>	<u>Proposed</u>
Assets	\$20,000	\$20,000
Debt	\$0	\$8,000
Equity	\$20,000	\$12,000
Debt/Equity ratio	0.00	2/3
Interest rate	n/a	8%
Shares outstanding	400	240
Share price	\$50	\$50

EPS and ROE Under Current Structure

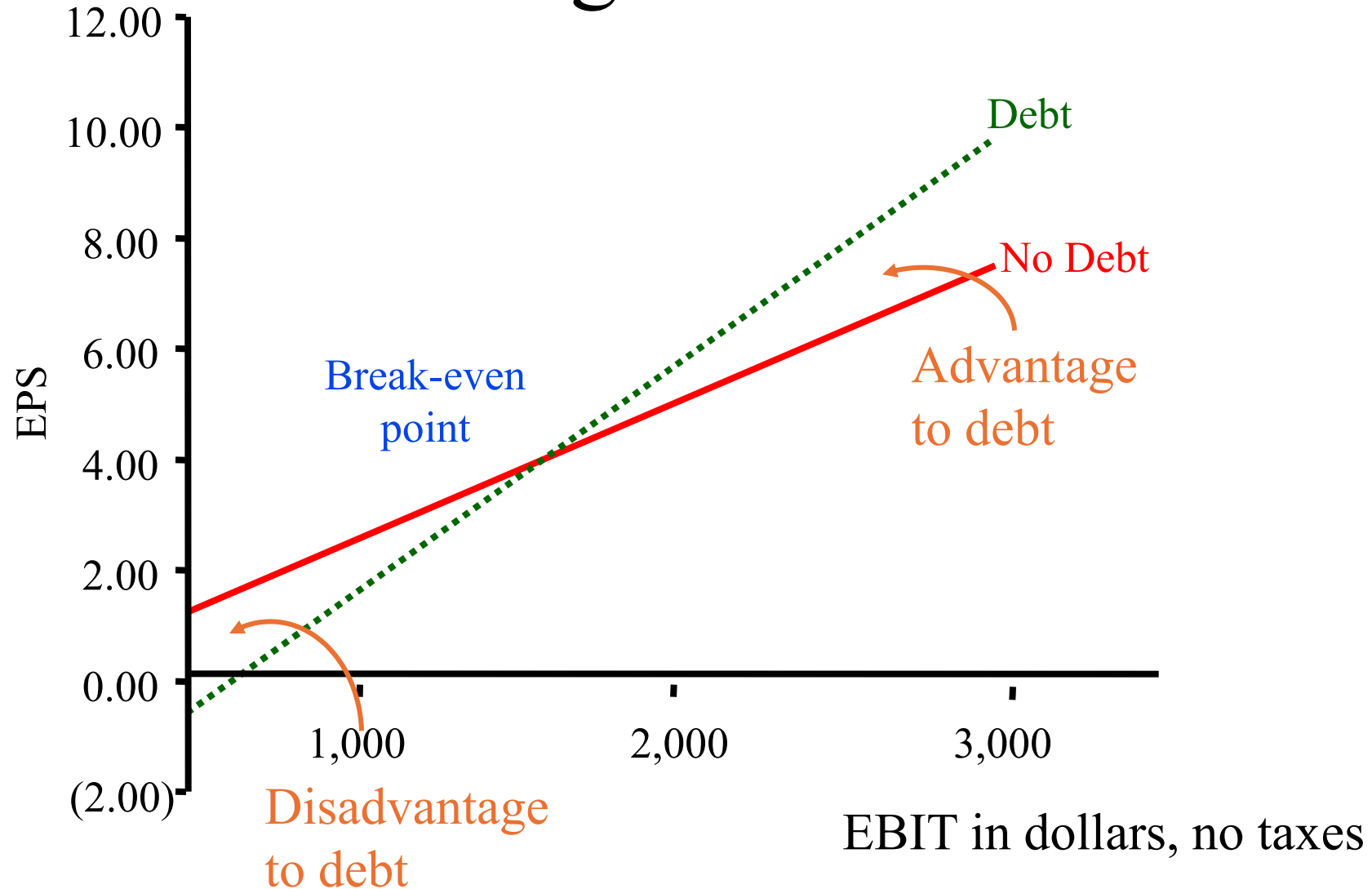
	<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
EBIT	\$1,000	\$2,000	\$3,000
Interest	0	0	0
<u>Net income</u>	<u>\$1,000</u>	<u>\$2,000</u>	<u>\$3,000</u>
EPS	\$2.50	\$5.00	\$7.50
ROA	5%	10%	15%
ROE	5%	10%	15%

Current Shares Outstanding = 400 shares

EPS and ROE Under Proposed Structure

	<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
EBIT	\$1,000	\$2,000	\$3,000
Interest	640	640	640
<u>Net income</u>	<u>\$360</u>	<u>\$1,360</u>	<u>\$2,360</u>
EPS	\$1.50	\$5.67	\$9.83
ROA	1.8%	6.8%	11.8%
ROE	3.0%	11.3%	19.7%
Proposed Shares Outstanding = 240 shares			

Financial Leverage and EPS



Assumptions of the M&M Model

- Homogeneous Expectations
- Homogeneous Business Risk Classes
- Perpetual Cash Flows
- Perfect Capital Markets:
 - Perfect competition
 - Firms and investors can borrow/lend at the same rate
 - Equal access to all relevant information
 - No transaction costs
 - No taxes

Homemade Leverage

- **STRATEGY A—BUY 24 SHARES OF THE LEVERED EQUITY**
- **STRATEGY B**
 1. Borrow \$800 from either a bank or, more likely, a brokerage house.
(If the brokerage house is the lender, we say that this activity is *going on margin* .)
 2. Use the borrowed proceeds plus your own investment of \$1,200 (a total of \$2,000) to buy 40 shares of the current unlevered equity at \$50 per share.

Example 1

Strategy A: BUY 24 SHARES OF THE LEVERED EQUITY (\$1200/\$50per share)

	<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
EPS	\$1.50	\$5.67	\$9.83

Of levered firm

Net profits	36	136	236
ROE	3.0%	11.3%	19.7%

$$=36/1200 \quad =136/1200=236/1200$$

Example 1

Strategy B:

	<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
<i>EPS of Unlevered Firm</i>	<i>\$2.50</i>	<i>\$5.00</i>	<i>\$7.50</i>
Earnings for 40 shares	\$100	\$200	\$300
<u>Less interest on \$800 (8%)</u>	<u>\$64</u>	<u>\$64</u>	<u>\$64</u>
Net Profits	\$36	\$136	\$236
<u>ROE (Net Profits / \$1,200)</u>	<u>3.0%</u>	<u>11.3%</u>	<u>19.7%</u>

We are buying 40 shares of a \$50 stock, using \$800 in margin.
We get the same ROE as if we bought into a levered firm.

Our personal debt-equity ratio is: $\frac{B}{S} = \frac{\$800}{\$1,200} = \frac{2}{3}$

Homemade (Un)Leverage

- **STRATEGY A—BUY 40 SHARES OF THE UNLEVERED EQUITY**
- **STRATEGY B**
 1. Buy 24 shares of the current levered equity at \$50 per share.
 2. lend \$800 to either a bank or, more likely, a brokerage house. (If the brokerage house is the borrower)

EPS and ROE Under Current Structure

Strategy A: A—BUY 40 SHARES OF THE UNLEVERED EQUITY

	<u>Expansion</u>	<u>Recession</u>	<u>Expected</u>
EPS	\$2.50	\$5.00	\$7.50
NET profits	100	200	300
ROE	5%	10%	15%
	$=100/2000$	$=200/2000$	$=300/2000$

Homemade (Un)Leverage: An Example

Strategy B:

	<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
<i>EPS of Levered Firm</i>	<i>\$1.50</i>	<i>\$5.67</i>	<i>\$9.83</i>
Earnings for 24 shares	\$36	\$136	\$236
Plus interest on \$800 (8%)	\$64	\$64	\$64
Net Profits	\$100	\$200	\$300
ROE (Net Profits / \$2,000)	5%	10%	15%

Buying 24 shares of an otherwise identical levered firm along with some of the firm's debt gets us to the ROE of the unlevered firm.

This is the fundamental insight of M&M

MM Proposition I (No Taxes)

- We can create a levered or unlevered position by adjusting the trading in our own account.
- This homemade leverage suggests that capital structure is irrelevant in determining the value of the firm:

$$V_L = V_U$$

MM Proposition II (No Taxes)

- Proposition II
 - Leverage increases the risk and return to stockholders

$$R_s = R_0 + (B / S) (R_0 - R_B)$$

R_B is the interest rate (cost of debt)

R_s is the return on (levered) equity (cost of equity)

R_0 is the return on unlevered equity (cost of capital)

B is the value of debt

S is the value of equity

MM Proposition II (No Taxes)

The derivation is straightforward:

$$R_{WACC} = \frac{B}{B+S} \times R_B + \frac{S}{B+S} \times R_S \quad \text{Then set } R_{WACC} = R_0$$

$$\frac{B}{B+S} \times R_B + \frac{S}{B+S} \times R_S = R_0 \quad \text{multiply both sides by } \frac{B+S}{S}$$

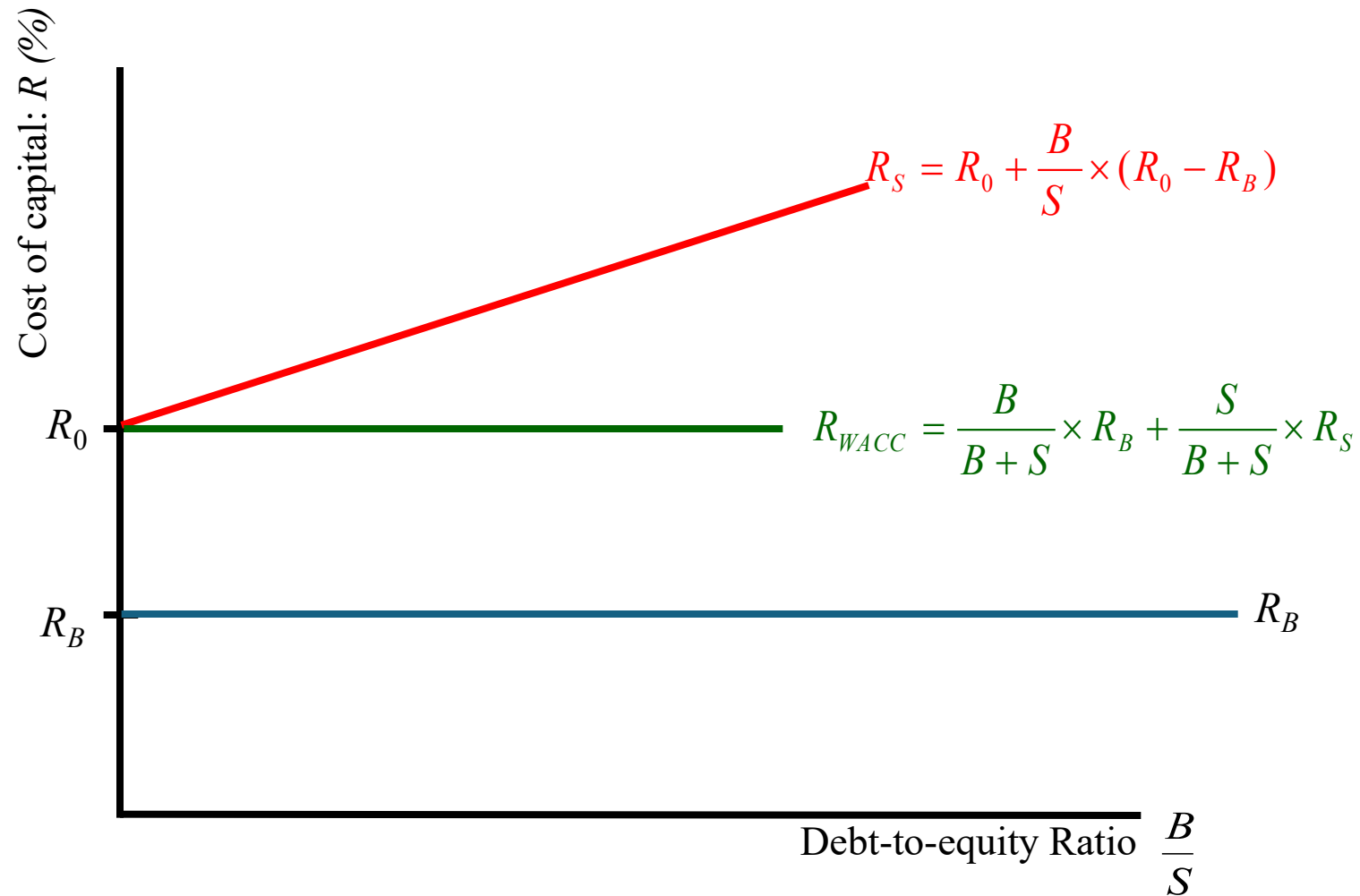
$$\frac{\cancel{B+S}}{S} \times \frac{B}{\cancel{B+S}} \times R_B + \frac{\cancel{B+S}}{S} \times \frac{\cancel{S}}{\cancel{B+S}} \times R_S = \frac{B+S}{S} R_0$$

$$\frac{B}{S} \times R_B + R_S = \frac{B+S}{S} R_0$$

$$\frac{B}{S} \times R_B + R_S = \frac{B}{S} R_0 + R_0$$

$$R_S = R_0 + \frac{B}{S} (R_0 - R_B)$$

MM Proposition II (No Taxes)



MM Proposition II (No Taxes)

Previous example continued:

$$R_0 = \text{EBIT} / \text{Unlevered equity} = \$2,000 / 20,000 = 10\%$$

$$\begin{aligned} R_s &= R_0 + (B / S) (R_0 - R_B) \\ &= 10\% + 2/3(10\% - 8\%) \\ &= 11.3\% \end{aligned}$$

$$R_{WACC} = \frac{S}{S+B} \times R_s + \frac{B}{S+B} \times R_B \times (1 - T_C)$$

Unlevered firm:

$$10\% = 20,000 / 20,000 \times 10\% + 0$$

Levered firm:

$$10\% = 12,000 / 20,000 \times 11.3\% + 8,000 / 20,000 \times 8\% \times (1 - 0)$$

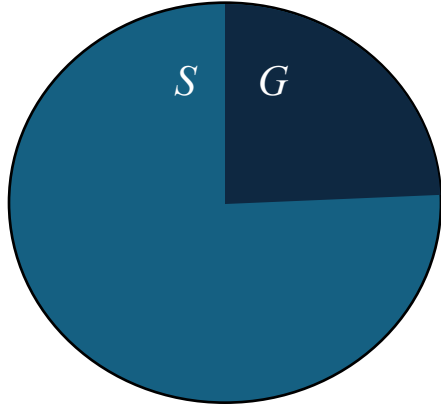
MM Proposition I (No Taxes)

- $V_U = \text{EBIT} / R_0$
 $= \$2000 / 10\% = \$20,000$
- $V_L = B + S$
 $= \$8000 + \text{NI} / R_S$
 $= \$8000 + 1360 / 11.3\%$
 $= \$20000$

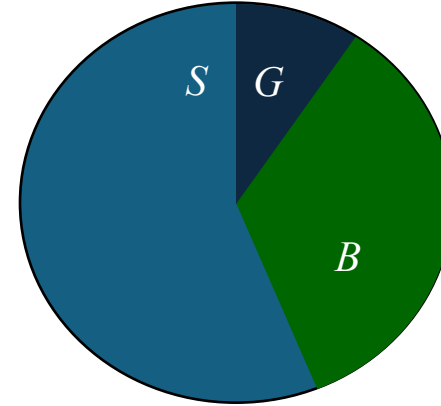
So, $V_L = V_U$

Pie model under corporate taxes

All-equity firm



Levered firm



The levered firm pays less in taxes than does the all-equity firm.

Thus, the sum of the debt plus the equity of the levered firm is greater than the equity of the unlevered firm.

This is how cutting the pie differently can make the pie “larger.”
-the government takes a smaller slice of the pie!

Total Cash Flow to Investors

		<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
All Equity	EBIT	\$1,000	\$2,000	\$3,000
	Interest	0	0	0
	EBT	\$1,000	\$2,000	\$3,000
	<u>Taxes ($t_c = 35\%$)</u>	<u>\$350</u>	<u>\$700</u>	<u>\$1,050</u>
	Total Cash Flow to S/H	\$650	\$1,300	\$1,950
		<u>Recession</u>	<u>Expected</u>	<u>Expansion</u>
Levered	EBIT	\$1,000	\$2,000	\$3,000
	Interest (\$8000 @ 8%)	640	640	640
	EBT	\$360	\$1,360	\$2,360
	<u>Taxes ($t_c = 35\%$)</u>	<u>\$126</u>	<u>\$476</u>	<u>\$826</u>
	Total Cash Flow	\$234+640	\$884+\$640	\$1,534+\$640
	(to both S/H & B/H):	<u>\$874</u>	\$1,524	\$2,174
	$EBIT(1-t_c)+t_cR_BB$	\$650+\$224	\$1,300+\$224	\$1,950+\$224
		<u>\$874</u>	\$1,524	\$2,174

MM Propositions I & II (With Taxes)

- Proposition I (with Corporate Taxes)

- Firm value increases with leverage

$$V_L = V_U + t_C B$$

- Proposition II (with Corporate Taxes)

- Some of the increase in equity risk and return is offset by the interest tax shield

$$R_S = R_0 + (B/S) \times (1 - t_C) \times (R_0 - R_B)$$

R_B is the interest rate (cost of debt)

R_S is the return on equity (cost of equity)

R_0 is the return on unlevered equity (cost of capital)

B is the value of debt

S is the value of equity

MM Proposition I (With Taxes)

The total cash flow to all stakeholders is

$$(EBIT - R_B B) \times (1 - t_C) + R_B B$$

The present value of this stream of cash flows is V_L

Clearly $(EBIT - R_B B) \times (1 - t_C) + R_B B =$

$$= EBIT \times (1 - t_C) - R_B B \times (1 - t_C) + R_B B$$

$$= EBIT \times (1 - t_C) - \cancel{R_B B} + R_B B t_C + \cancel{R_B B}$$

The present value of the first term is V_U

The present value of the second term is $t_C B$

$$\therefore V_L = V_U + t_C B$$

MM Proposition II (With Taxes)

Start with M&M Proposition I with taxes: $V_L = V_U + t_C B$

Since $V_L = S + B \Rightarrow S + B = V_U + t_C B$

$$V_U = S + B(1 - t_C)$$

The cash flows from each side of the balance sheet must equal:

$$SR_S + BR_B = V_U R_0 + t_C BR_B$$

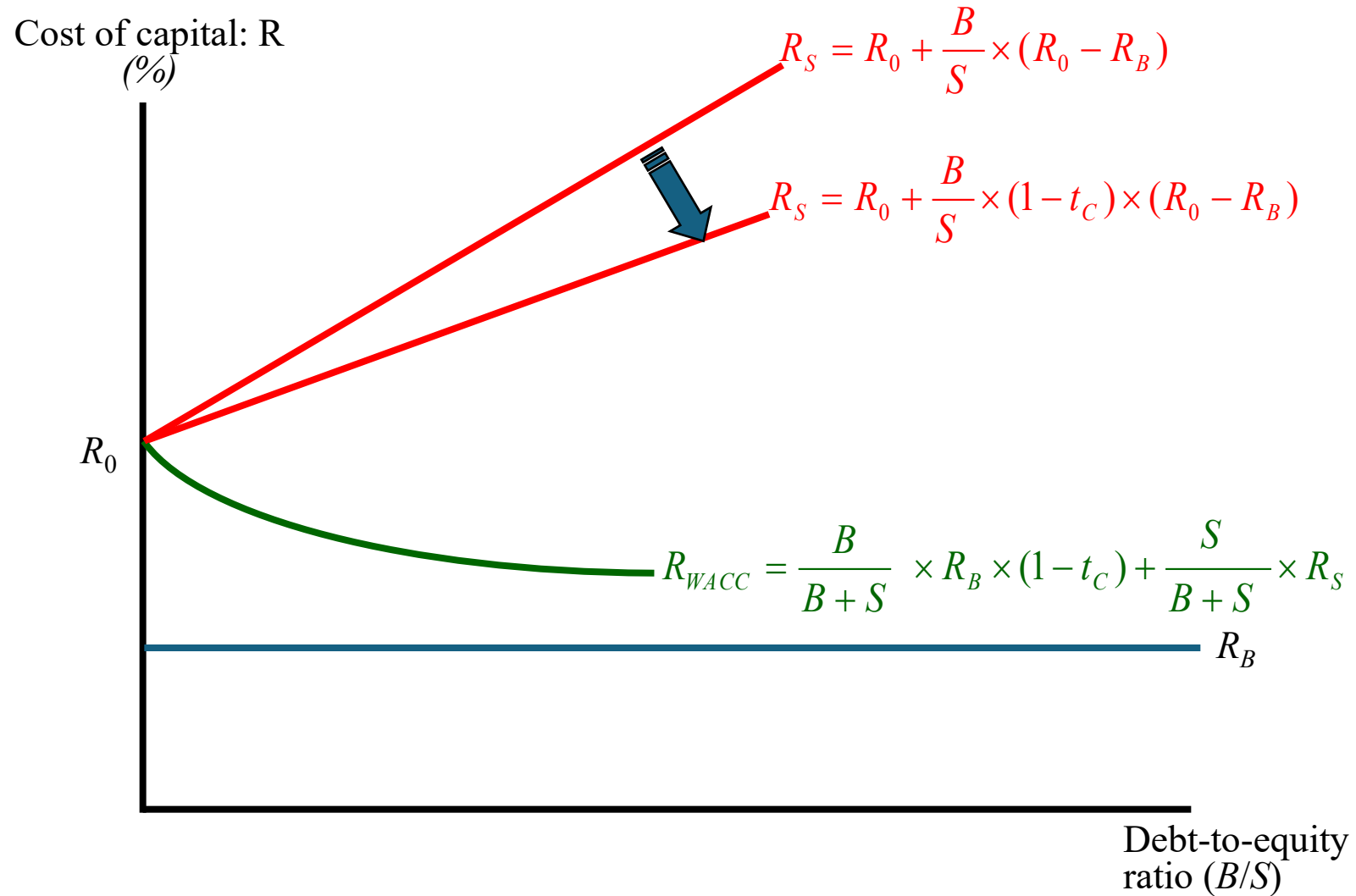
$$SR_S + BR_B = [S + B(1 - t_C)]R_0 + t_C R_B B$$

Divide both sides by S

$$R_S + \frac{B}{S} R_B = [1 + \frac{B}{S} (1 - t_C)]R_0 + \frac{B}{S} t_C R_B$$

Which quickly reduces to $R_S = R_0 + \frac{B}{S} \times (1 - t_C) \times (R_0 - R_B)$

The Effect of Financial Leverage



The Effect of Financial Leverage

Previous example continued:

$$\begin{aligned} V_L &= V_U + t_C B \\ &= \$1,300/0.1 + 35\% \times 8000 = \$15,800 \end{aligned}$$

$$S = V_L - B = \$15,800 - 8,000 = \$7,800$$

$$\begin{aligned} R_S &= R_0 + (B/S) \times (1 - t_C) \times (R_0 - R_B) \\ &= 10\% + 8000/7800 \times (1 - 35\%) \times (10\% - 8\%) \\ &= 11.33\% \end{aligned}$$

The Effect of Financial Leverage

Previous example continued:

$$R_{WACC} = \frac{S}{S+B} \times R_S + \frac{B}{S+B} \times R_B \times (1 - T_C)$$

$$= 7,800/15,800 \times 11.33\% + 8,000/15,800 \times 8\% \times (1 - 35\%)$$

$$= 8.23\%$$

$$V_L = EBIT(1 - T_C) / RWACC = 2,000 \times (1 - 35\%) / 8.23\% = 15,800$$

Summary: No Taxes

- In a world of no taxes, the value of the firm is unaffected by capital structure.
- This is M&M Proposition I:

$$V_L = V_U$$

- Proposition I holds because shareholders can achieve any pattern of payouts they desire with homemade leverage.
- In a world of no taxes, M&M Proposition II states that leverage increases the risk and return to stockholders in proportion resulting to no change in the net value of the firm.

$$R_S = R_0 + \frac{B}{S} \times (R_0 - R_B)$$

Summary: Taxes

- ▶ In a world of taxes, but no bankruptcy costs, the value of the firm increases with leverage.
- ▶ This is M&M Proposition I:

$$V_L = V_U + t_C B$$

- ▶ Proposition I holds because shareholders can achieve any pattern of payouts they desire with homemade leverage.
- ▶ In a world of taxes, M&M Proposition II states that leverage increases the risk and return to stockholders.

$$R_S = R_0 + \frac{B}{S} \times (1 - t_C) \times (R_0 - R_B)$$



4. **Break-Even EBIT** Hale Corporation is comparing two different capital structures, an all-equity plan (Plan I) and a levered plan (Plan II). Under Plan I, the company would have 145,000 shares of stock outstanding. Under Plan II, there would be 90,000 shares of stock outstanding and \$3,047,000 in debt outstanding. The interest rate on the debt is 8 percent and there are no taxes.
- a. If EBIT is \$400,000, which plan will result in the higher EPS?
 - b. If EBIT is \$700,000, which plan will result in the higher EPS?
 - c. What is the break-even EBIT?

4. *a.* Under Plan I, the unlevered company, net income is the same as EBIT with no corporate tax. The EPS under this capitalization will be:

$$\text{EPS} = \$400,000 / 145,000 \text{ shares}$$

$$\text{EPS} = \$2.76$$

Under Plan II, the levered company, EBIT will be reduced by the interest payment. The interest payment is the amount of debt times the interest rate, so:

$$\text{NI} = \$400,000 - .08(\$3,047,000)$$

$$\text{NI} = \$156,240$$

And the EPS will be:

$$\text{EPS} = \$156,240 / 90,000 \text{ shares}$$

$$\text{EPS} = \$1.74$$

Plan I has the higher EPS when EBIT is \$400,000.

b. Under Plan I, the net income is \$700,000 and the EPS is:

$$\text{EPS} = \$700,000 / 145,000 \text{ shares}$$

$$\text{EPS} = \$4.83$$

Under Plan II, the net income is:

$$\text{NI} = \$700,000 - .08(\$3,047,000)$$

$$\text{NI} = \$456,240$$

And the EPS is:

$$\text{EPS} = \$456,240 / 90,000 \text{ shares}$$

$$\text{EPS} = \$5.07$$

Plan II has the higher EPS when EBIT is \$700,000.

c. To find the breakeven EBIT for two different capital structures, we set the equations for EPS equal to each other and solve for EBIT. The breakeven EBIT is:

$$\text{EBIT} / 145,000 = [\text{EBIT} - .08(\$3,047,000)] / 90,000$$

$$\text{EBIT} = \$642,640$$

- 9. Homemade Leverage and WACC** ABC Co. and XYZ Co. are identical firms in all respects except for their capital structures. ABC is all equity financed with \$650,000 in stock. XYZ uses both stock and perpetual debt; its stock is worth \$325,000 and the interest rate on its debt is 8 percent. Both firms expect EBIT to be \$70,000. Ignore taxes.
- a. Richard owns \$20,000 worth of XYZ's stock. What rate of return is he expecting?
 - b. Show how Richard could generate exactly the same cash flows and rate of return by investing in ABC and using homemade leverage.
 - c. What is the cost of equity for ABC? What is it for XYZ?
 - d. What is the WACC for ABC? For XYZ? What principle have you illustrated?

9. *a.* The rate of return earned will be the dividend yield. The company has debt, so it must make an interest payment. The net income for the company is:

$$NI = \$70,000 - .08(\$325,000)$$

$$NI = \$44,000$$

The investor will receive dividends in proportion to the percentage of the company's share they own. The total dividends received by the shareholder will be:

$$\text{Dividends received} = \$44,000(\$20,000/\$325,000)$$

$$\text{Dividends received} = \$2,708$$

So the return the shareholder expects is:

$$R = \$2,708/\$20,000$$

$$R = .1354, \text{ or } 13.54\%$$

b. To generate exactly the same cash flows in the other company, the shareholder needs to match the capital structure of ABC. The shareholder should sell all shares in XYZ. This will net \$20,000. The shareholder should then borrow \$20,000. This will create an interest cash flow of:

$$\text{Interest cash flow} = .08(\$20,000)$$

$$\text{Interest cash flow} = \$1,600$$

The investor should then use the proceeds of the stock sale and the loan to buy shares in ABC. The investor will receive dividends in proportion to the percentage of the company's shares they own. The total dividends received by the shareholder will be:

$$\text{Dividends received} = \$70,000(\$40,000/\$650,000)$$

$$\text{Dividends received} = \$4,308$$

The total cash flow for the shareholder will be:

$$\text{Total cash flow} = \$4,308 - 1,600$$

$$\text{Total cash flow} = \$2,708$$

The shareholder's return in this case will be:

$$R = \$2,708/\$20,000$$

$$R = .1354, \text{ or } 13.54\%$$

c. ABC is an all equity company, so:

$$R_S = R_0 = \$70,000/\$650,000$$

$$R_S = .1077, \text{ or } 10.77\%$$

To find the cost of equity for XYZ, we need to use M&M Proposition II, so:

$$R_S = R_0 + (R_0 - R_B)(B/S)(1 - t_C)$$

$$R_S = .1077 + (.1077 - .08)(1)(1)$$

$$R_S = .1354, \text{ or } 13.54\%$$

d. To find the WACC for each company, we need to use the WACC equation:

$$\text{WACC} = (S/V)R_S + (D/V)R_D(1 - t_C)$$

So, for ABC, the WACC is:

$$\text{WACC} = (1)(.1077) + (0)(.08)$$

$$\text{WACC} = .1077, \text{ or } 10.77\%$$

And for XYZ, the WACC is:

$$\text{WACC} = (1/2)(.1354) + (1/2)(.08)$$

$$\text{WACC} = .1077, \text{ or } 10.77\%$$

- 12. Calculating WACC** Weston Industries has a debt–equity ratio of 1.4. Its WACC is 8.3 percent, and its cost of debt is 5.7 percent. The corporate tax rate is 35 percent.
- a. What is the company's cost of equity capital?
 - b. What is the company's unlevered cost of equity capital?
 - c. What would the cost of equity be if the debt–equity ratio were 2? What if it were 1? What if it were zero?

- 12.** a. With the information provided, we can use the equation for calculating WACC to find the cost of equity. The equation for WACC is:

$$\text{WACC} = (S/V)R_S + (B/V)R_B(1 - t_C)$$

The company has a debt-equity ratio of 1.4, which implies the weight of debt is $1.4/2.4$, and the weight of equity is $1/2.4$, so

$$\begin{aligned}\text{WACC} &= .083 = (1/2.4)R_S + (1.4/2.4)(.057)(1 - .35) \\ R_S &= .1473, \text{ or } 14.73\%\end{aligned}$$

b. To find the unlevered cost of equity we need to use M&M Proposition II with taxes, so:

$$\begin{aligned}R_S &= R_0 + (R_0 - R_B)(B/S)(1 - t_C) \\ .1473 &= R_0 + (R_0 - .057)(1.4)(1 - .35) \\ R_0 &= .1043, \text{ or } 10.43\%\end{aligned}$$

c. To find the cost of equity under different capital structures, we can again use M&M Proposition II with taxes. With a debt-equity ratio of 2, the cost of equity is:

$$\begin{aligned}R_S &= R_0 + (R_0 - R_B)(B/S)(1 - t_C) \\ R_S &= .1043 + (.1043 - .057)(2)(1 - .35) \\ R_S &= .1658, \text{ or } 16.58\%\end{aligned}$$

- 18. Firm Value** Full Moon Corporation expects an EBIT of \$27,750 every year forever. The company currently has no debt, and its cost of equity is 12.8 percent. The corporate tax rate is 35 percent.
- a. What is the current value of the company?
 - b. Suppose the company can borrow at 7.1 percent. What will the value of the firm be if the company takes on debt equal to 50 percent of its unlevered value? What if it takes on debt equal to 100 percent of its unlevered value?
 - c. What will the value of the firm be if the company takes on debt equal to 50 percent of its levered value? What if the company takes on debt equal to 100 percent of its levered value?

- 18.** a. With no debt, we are finding the value of an unlevered firm, so:

$$V_U = \text{EBIT}(1 - t_C)/R_0$$

$$V_U = \$27,750(1 - .35)/.128$$

$$V_U = \$140,917.97$$

b. The general expression for the value of a leveraged firm is:

$$V_L = V_U + t_C B$$

If debt is 50 percent of V_U , then $B = (.50)V_U$, and we have:

$$V_L = V_U + t_C[(.50)V_U]$$

$$V_L = \$140,917.97 + .35(.50)(\$140,917.97)$$

$$V_L = \$165,578.61$$

And if debt is 100 percent of V_U , then $B = (1.0)V_U$, and we have:

$$V_L = V_U + t_C[(1.0)V_U]$$

$$V_L = \$140,917.97 + .35(1.0)(\$140,917.97)$$

$$V_L = \$190,239.26$$

c. According to M&M Proposition I with taxes:

$$V_L = V_U + t_C B$$

With debt being 50 percent of the value of the levered firm, D must equal $(.50)V_L$, so:

$$V_L = V_U + t_C[(.50)V_L]$$

$$V_L = \$140,917.97 + .35(.50)(V_L)$$

$$V_L = \$170,809.66$$

- 20. MM Proposition I without Taxes** Alpha Corporation and Beta Corporation are identical in every way except their capital structures. Alpha Corporation, an all-equity firm, has 19,000 shares of stock outstanding, currently worth \$23 per share. Beta Corporation uses leverage in its capital structure. The market value of Beta's debt is \$85,000, and its cost of debt is 7 percent. Each firm is expected to have earnings before interest of \$71,000 in perpetuity. Neither firm pays taxes. Assume that every investor can borrow at 7 percent per year.
- a. What is the value of Alpha Corporation?
 - b. What is the value of Beta Corporation?
 - c. What is the market value of Beta Corporation's equity?
 - d. How much will it cost to purchase 20 percent of each firm's equity?
 - e. Assuming each firm meets its earnings estimates, what will be the dollar return to each purchase in part (d) over the next year?
 - f. Construct an investment strategy in which an investor purchases 20 percent of Alpha's equity and replicates both the cost and dollar return of purchasing 20 percent of Beta's equity.
 - g. Is Alpha's equity more or less risky than Beta's equity? Explain.

20. *a.* Since Alpha Corporation is an all-equity firm, its value is equal to the market value of its outstanding shares. Alpha has 19,000 shares of common stock outstanding, worth \$23 per share, so the value of Alpha Corporation is:

$$V_{\text{Alpha}} = 19,000(\$23)$$

$$V_{\text{Alpha}} = \$437,000$$

- b.* Modigliani-Miller Proposition I states that in the absence of taxes, the value of a levered firm equals the value of an otherwise identical unlevered firm. Since Beta Corporation is identical to Alpha Corporation in every way except its capital structure and neither firm pays taxes, the value of the two firms should be equal. So, the value of Beta Corporation is \$437,000 as well.

- c.* The value of a levered firm equals the market value of its debt plus the market value of its equity. So, the value of Beta's equity is:

$$V_L = B + S$$

$$\$437,000 = \$85,000 + S$$

$$S = \$352,000$$

d. The investor would need to invest 20 percent of the total market value of Alpha's equity, which is:

$$\text{Amount to invest in Alpha} = .20(\$437,000)$$

$$\text{Amount to invest in Alpha} = \$87,400$$

Beta has less equity outstanding, so to purchase 20 percent of Beta's equity, the investor would need:

$$\text{Amount to invest in Beta} = .20(\$352,000)$$

$$\text{Amount to invest in Beta} = \$70,400$$

e. Alpha has no interest payments, so the dollar return to an investor who owns 20 percent of the company's equity would be:

$$\text{Dollar return on Alpha investment} = .20(\$71,000)$$

$$\text{Dollar return on Alpha investment} = \$14,200$$

Beta Corporation has an interest payment due on its debt in the amount of:

$$\text{Interest on Beta's debt} = .07(\$85,000)$$

$$\text{Interest on Beta's debt} = \$5,950$$

So, the investor who owns 20 percent of the company would receive 20 percent of EBIT minus the interest expense, or:

$$\text{Dollar return on Beta investment} = .20(\$71,000 - 5,950)$$

$$\text{Dollar return on Beta investment} = \$13,010$$

f. From part *d*, we know the initial cost of purchasing 20 percent of Alpha Corporation's equity is \$87,400, but the cost to an investor of purchasing 20 percent of Beta Corporation's equity is only \$70,400. In order to purchase \$87,400 worth of Alpha's equity using only \$70,400 of his own money, the investor must borrow \$17,000 to cover the difference. The investor will receive the same dollar return from the Alpha investment, but will pay interest on the amount borrowed, so the net dollar return to the investment is:

$$\text{Net dollar return} = \$14,200 - .07(\$17,000)$$

$$\text{Net dollar return} = \$13,010$$

Notice that this amount exactly matches the dollar return to an investor who purchases 20 percent of Beta's equity.

g. The equity of Beta Corporation is riskier. Beta must pay off its debt holders before its equity holders receive any of the firm's earnings. If the firm does not do particularly well, all of the firm's earnings may be needed to repay its debt holders, and equity holders will receive nothing.

- 23. MM Propositions** Martin Corporation is planning to repurchase part of its common stock by issuing corporate debt. As a result, the firm's debt–equity ratio is expected to rise from 20 percent to 35 percent. The firm currently has \$4.96 million worth of debt outstanding. The cost of this debt is 7 percent per year. The firm expects to have EBIT of \$2.92 million per year in perpetuity and pays no taxes.
- a. What is the market value of the firm before and after the repurchase announcement?
 - b. What is the expected return on the firm's equity before the announcement of the stock repurchase plan?
 - c. What is the expected return on the equity of an otherwise identical all-equity firm?
 - d. What is the expected return on the firm's equity after the announcement of the stock repurchase plan?

23. a. we can find that the value of the outstanding equity must be:

$$\text{Debt-equity ratio} = B/S$$

$$.20 = \$4,960,000/S$$

$$S = \$24,800,000$$

The value of a levered firm is equal to the sum of the market value of the firm's debt and the market value of the firm's equity, so:

$$V_L = B + S$$

$$V_L = \$4,960,000 + 24,800,000$$

$$V_L = \$29,760,000$$

- b. The expected return on a firm's equity is the ratio of annual earnings to the market value of the firm's equity, or return on equity. Before the restructuring, the company was expected to pay interest in the amount of:

$$\text{Interest payment} = .07(\$4,960,000)$$

$$\text{Interest payment} = \$347,200$$

The return on equity, which is equal to R_S , will be:

$$\text{ROE} = R_S = (\$2,920,000 - 347,200)/\$24,800,000$$

$$R_S = .1037, \text{ or } 10.37\%$$

- c. According to Modigliani-Miller Proposition II with no taxes:

$$R_S = R_0 + (B/S)(R_0 - R_B)$$

$$.1037 = R_0 + (.20)(R_0 - .07)$$

$$R_0 = .0981, \text{ or } 9.81\%$$

This problem can also be solved in the following way:

$$R_0 = \text{Earnings before interest}/V_U$$

$$R_0 = \$2,920,000/\$29,760,000$$

$$R_0 = .0981, \text{ or } 9.81\%$$

d. In part c, we calculated the cost of an all-equity firm. We can use Modigliani-Miller Proposition II with no taxes again to find the cost of equity for the firm with the new leverage ratio. The cost of equity under the stock repurchase plan will be:

$$R_S = R_0 + (B/S)(R_0 - R_B)$$

$$R_S = .0981 + (.35)(.0981 - .07)$$

$$R_S = .1080, \text{ or } 10.80\%$$

24. Stock Value and Leverage Green Manufacturing, Inc., plans to announce that it will issue \$1.6 million of perpetual debt and use the proceeds to repurchase common stock. The bonds will sell at par with

an annual coupon rate of 6 percent. The company is currently an all-equity firm worth \$7.7 million with 260,000 shares of common stock outstanding. After the sale of the bonds, the company will maintain the new capital structure indefinitely. The company currently generates annual pretax earnings of \$1.86 million. This level of earnings is expected to remain constant in perpetuity. The corporate tax rate is 40 percent.

- a. What is the expected return on equity before the announcement of the debt issue?
- b. Construct the market value balance sheet before the announcement of the debt issue. What is the price per share of the firm's equity?
- c. Construct the market value balance sheet immediately after the announcement of the debt issue.
- d. What is the stock price per share immediately after the repurchase announcement?
- e. How many shares will the company repurchase as a result of the debt issue? How many shares of common stock will remain after the repurchase?
- f. Construct the market value balance sheet after the restructuring.
- g. What is the required return on the company's equity after the restructuring?

- 24.** *a.* The expected return on a firm's equity is the ratio of annual aftertax earnings to the market value of the firm's equity. The amount the firm must pay each year in taxes will be:

$$\text{Taxes} = .40(\$1,860,000)$$

$$\text{Taxes} = \$744,000$$

So, the return on the unlevered equity will be:

$$R_0 = (\$1,860,000 - 744,000)/\$7,700,000$$

$$R_0 = .1449, \text{ or } 14.49\%$$

Notice that perpetual annual earnings of \$1,116,000 (= \$1,860,000 – 744,000), discounted at 14.49 percent, yields the market value of the firm's equity

b. The company's market value balance sheet before the announcement of the debt issue is:

			Debt	0
	Assets	<u>\$7,700,000</u>	Equity	<u>\$7,700,000</u>
	Total assets	\$7,700,000	Total D&E	\$7,700,000

The price per share is the total market value of the stock divided by the shares outstanding, or:

$$\text{Price per share} = \$7,700,000 / 260,000$$

$$\text{Price per share} = \$29.62$$

c. Modigliani-Miller Proposition I states that in a world with corporate taxes:

$$V_L = V_U + t_C B$$

When the company announces the debt issue, the value of the firm will increase by the present value of the tax shield on the debt. The present value of the tax shield is:

$$\text{PV(Tax Shield)} = t_C B$$

$$\text{PV(Tax Shield)} = .40(\$1,600,000)$$

$$\text{PV(Tax Shield)} = \$640,000$$

Therefore, the value of the company will increase by \$640,000 as a result of the debt issue.

The value of the company after the repurchase announcement is:

$$V_L = V_U + t_C B$$

$$V_L = \$7,700,000 + .40(\$1,600,000)$$

$$V_L = \$8,340,000$$

Since the firm has not yet issued any debt, the company's equity is also worth \$8,340,000.

The company's market value balance sheet after the announcement of the debt issue is:

	Old assets	\$7,700,000	Debt	0
	PV(tax shield)	<u>640,000</u>	Equity	<u>\$8,340,000</u>
	Total assets	\$8,340,000	Total D&E	\$8,340,000

d. The share price immediately after the announcement of the debt issue will be:

$$\text{New share price} = \$8,340,000 / 260,000$$

$$\text{New share price} = \$32.08$$

e. The number of shares repurchased will be the amount of the debt issue divided by the new share price, or:

$$\text{Shares repurchased} = \$1,600,000 / \$32.08$$

$$\text{Shares repurchased} = 49,880.10$$

The number of shares outstanding will be the current number of shares minus the number of shares repurchased, or:

$$\text{New shares outstanding} = 260,000 - 49,880.10$$

$$\text{New shares outstanding} = 210,119.90$$

- f. The share price will remain the same after restructuring takes place.
The total market value of the outstanding equity in the company will be:

Market value of equity = $\$32.08(210,119.90)$

Market value of equity = $\$6,740,000$

The market-value balance sheet after the restructuring is:

	Old assets	\$7,700,000	Debt	\$1,600,000
	PV(tax shield)	<u>640,000</u>	Equity	<u>6,740,000</u>
	Total assets	\$8,340,000	Total D&E	\$8,340,000

- g. According to Modigliani-Miller Proposition II with corporate taxes

$$R_S = R_0 + (B/S)(R_0 - R_B)(1 - t_C)$$

$$R_S = .1449 + (\$1,600,000/\$6,740,000)(.1449 - .06)(1 - .40)$$

$$R_S = .1570, \text{ or } 15.70\%$$